

AI PAYS OFF FROM THE **INSIDE** OUT

Most AI pilots in a plant fail on the outer layers, at roll-out and scale, because the inner layers were never built. Build from the centre outward. Each layer is what the next one stands on, so **don't start a layer until the one inside it holds**.

1 Mindset · Start from the problem, not the tool

THIS WAY SCALES

Write the problem as a measured gap before naming any technology. Ask: would we fix this without AI?

THIS WAY STALLS

Tool first: buy a platform, then hunt for a use case.

OUTPUT

A problem statement with a baseline number and one owner.

2 Process · Stabilise before you automate

THIS WAY SCALES

Walk the process as it really runs, not as the SOP says. Remove the obvious waste and variation, then standardise the best-known way.

THIS WAY STALLS

Automating the mess: the model learns today's variation and repeats it faster.

OUTPUT

A stable process with standard work and a baseline you trust.

3 Data · Give every data set a named owner

THIS WAY SCALES

Name an owner for each source the use case needs. Fix quality where data is created: the batch record, sensor or form.

THIS WAY STALLS

The data swamp: everything collected, nobody accountable, one clean-up, then stale again.

OUTPUT

A data map: source, owner, quality check, refresh rhythm.

4 People · Build it with the people who run it

THIS WAY SCALES

Put operators and supervisors in the design team from week one. Teach the why: what the tool sees, what it can't, and when to override it.

THIS WAY STALLS

Pushed from HQ: the line keeps its shadow Excel, and nobody opens the tool.

OUTPUT

Named frontline co-designers and a one-page "when to trust it" guide.

5 Scale · Prove it, then spread it

THIS WAY SCALES

Measure the result against the layer 1 baseline, not the pilot plan. Spread it through Yokoten: share the why, let each line adapt.

THIS WAY STALLS

Pilot purgatory, and vanity metrics like "models deployed".

OUTPUT

A before/after result signed by the process owner, and a "where else?" list.



WHERE IS YOUR PROJECT REALLY STUCK?

What you hear in the review meeting is a symptom. The cause is usually a layer nearer the centre. Find the layer, then make the first move there.

WHAT YOU HEAR	MISSING LAYER	FIRST MOVE
"The pilot worked, but nobody uses it."	People	Run one redesign session with two operators from the line this week.
"The model is accurate, but nothing improved."	Mindset	Restate the problem as a measured gap. If there is no gap, stop.
"We need more data first."	Data	Name owners for the data you already have before collecting more.
"Results swing from shift to shift."	Process	Standardise the work before you retrain the model.
"It worked at one site and flopped at the next."	Scale	Yokoten the why and the standard, not just the file.
"We can't show the return."	Mindset	There was no baseline. Measure one now, before the next change.

GATE CHECK BEFORE YOU MOVE OUTWARD

LAYER	ANSWER YES BEFORE THE NEXT LAYER
1 · Mindset	Can we state the gap as a number, with an owner?
2 · Process	Has the process held its baseline without firefighting?
3 · Data	Does every source have an owner and a quality check?
4 · People	Did operators shape it, and do they know when to override it?
5 · Scale	Is the gain measured against the baseline and signed off?

MAKE IT STICK

- **One problem, one owner.** A use case with three sponsors has none.
- **Baseline before change.** Without it, every result is an opinion.
- **Operators in the room** from the first design session, not the go-live.
- **AI structures the evidence; people decide.** Write down who can override it.
- **Spread the why, not the file.** Each line adapts it to its own process.

SELF-CHECK: SIGNS YOU ARE BUILDING FROM THE OUTSIDE IN

- ☐ The vendor was chosen before the problem was written
- ☐ Nobody can quote the baseline
- ☐ The process changes every few weeks
- ☐ Data is "owned by IT"
- ☐ Operators first saw the tool at go-live
- ☐ The line still runs its own spreadsheet
- ☐ Success is reported as "models deployed"
- ☐ The pilot has been "nearly ready to scale" for months

Three or more ticks? Pause the roll-out and go back to the lowest layer you ticked.



YOUR INSIDE-OUT SCORECARD

Score one AI or digital project with the people who run the process. Be honest: a 2 you can fix beats a 4 you can't prove.

THE PROJECT

THE GAP (TODAY → TARGET)

SCORE EACH LAYER, CENTRE FIRST					
LAYER	WHERE WE ARE TODAY	EVIDENCE	SCORE	NEXT MOVE	OWNER
1 • Mindset			1 2 3 4 5		
2 • Process			1 2 3 4 5		
3 • Data			1 2 3 4 5		
4 • People			1 2 3 4 5		
5 • Scale			1 2 3 4 5		

WHERE TO START

Find the **lowest score nearest the centre**. That is your next move, even if an outer layer scores lower. Fixing scale while the process still moves only spreads the problem.

BASELINE (LAYER 1)			WHERE ELSE? (LAYER 5)		
MEASURE	TODAY	TARGET	LINE / SITE	SAME PROBLEM?	OWNER

RE-SCORE IN 30 DAYS					
DATE	MINDSET	PROCESS	DATA	PEOPLE	SCALE